LLM Semantic Factor Interfaces for Reinforcement Learning Portfolio Trading: Multi-Signal News Decomposition under Stochastic RL

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Abstract

Sequential agents increasingly encounter *sparse* unstructured text: informative observations arrive on a minority of steps, while most transitions carry only a neutral default. We ask how such text should enter decision state—as dense embeddings, scalars, or a small set of interpretable coordinates—and how to separate representation quality from optimisation behaviour when LLMs supply the encoding. We instantiate **Semantic State Abstraction Interfaces** (SSAI): prompted LLM maps from documents to K auditable axes with neutral defaults on no-news days (Definition 1, Appendix A), evaluated here with $K{=}4$ (*sentiment*, *risk*, *confidence*, *volatility forecast*) in a portfolio setting (**FinRL-MultiSignal**). Direct portfolios (SFP/SRF/SCW), supervised ridge forecasts, and paired RL agents (DP-PPO, SAC) share the same SSAI so representation effects can be read off independently of any single optimiser. On a 30-stock NASDAQ-100 panel (train 2013–2018; test 2019–2023), four-factor SFP reaches **307.2%** cumulative return and Sharpe **1.067** vs. **291.9%/1.050** sentiment-only and **243.6%/1.032** equal-weight buy-and-hold; residualizing non-sentiment axes preserves gains; conviction weighting reaches **314.1%**. Naively appending sparse LLM semantic columns hurts a dense price/technical ridge forecaster, while lexical headline aggregates (VADER; TF–IDF/SVD) and a validation-selected high-conviction semantic tilt recover Sharpe near the price-only ridge frontier—consistent with SSAI as a *sparse allocation interface* rather than a naive dense LLM feature block. The RL block is intentionally diagnostic: DP-PPO trails buy-and-hold but shows masking sensitivity; SAC with identical SSAI observations nearly matches buy-and-hold return and improves Sharpe (**1.128** vs. **1.032**), isolating algorithm dependence rather than claiming a production trading system. Across eight DP-PPO seeds, full scoring improves mean return (193.7%$\pm$43.7% vs. 187.2%$\pm$42.9%) and Sharpe (0.913$\pm$0.094 vs. 0.889$\pm$0.084) versus neutral-masked evaluation of the same checkpoints (bootstrap Sharpe CIs $[0.85, 0.98]$ vs. $[0.83, 0.94]$), but paired Wilcoxon tests remain non-significant ($p{\approx}0.25$–0.31). Overall, LLM-derived semantic factorisation functions as a reproducible SSAI whose strongest evidence is direct-portfolio and supervised sparse-use behaviour; RL outcomes must be read as optimisation stress tests under heavy neutral-default regimes.

1 Introduction

Deep Reinforcement Learning (DRL) is widely studied for sequential decision problems where observations mix numerical signals with unstructured evidence [Mnih et al., 2015, Schulman et al.,

40th Conference on Neural Information Processing Systems (NeurIPS 2026).

2017, Liu et al., 2021]. Large language models (LLMs) offer flexible encoders for text [Brown et al., 2020, Liu et al., 2023, Wu et al., 2023], yet most pipelines obscure a basic design question: should textual evidence appear as a high-dimensional latent vector, as a single scalar (e.g., sentiment), or as a small inventory of named semantic coordinates—especially when text arrives only on a sparse subset of decision dates?

Prior work has connected NLP with RL trading in limited ways. Yang et al. [2020a] use technical indicators as the sole state representation; Benhenda [2025] add a single sentiment score from DeepSeek-V3; and Lopez-Lira and Tang [2023] use ChatGPT to classify daily market sentiment as a binary signal. A single sentiment score is information-sparse: it conflates orthogonal concepts—whether a headline is positive differs fundamentally from whether the underlying situation is *risky* or whether the market is likely to be *volatile*.

Positioning. We adopt *Semantic State Abstraction Interfaces* (SSAIs), formally defined in Appendix A: maps ϕ from document sets $\mathcal{N}_{s,d}$ to K interpretable coordinates elicited by fixed prompts from a frozen LLM, with a neutral default when $\mathcal{N}_{s,d} = \emptyset$. Methodologically, the contribution is not a new RL gradient formula but (i) a reusable SSAI template—auditable axes, sparse-by-design neutral fills, shared ϕ across estimators—and (ii) an evaluation protocol that separates representation hypotheses from optimisation variance: direct factor portfolios and supervised dense-vs-sparse regressions read off semantic signal with minimal policy entropy; DP-PPO and SAC supply paired nonlinear stress tests on identical SSAI observations.

Trading serves as a demanding instantiation because aggregate outcomes are noisy, textual coverage is uneven (14.8% non-neutral stock-days), and dense ridge augmentation provides an explicit "wrong" use of sparse semantics against which SSAI tilts and factor portfolios contrast.

Research questions. We organise the experiments around three concrete questions: **RQ1**: do multi-axis LLM signals change policy behaviour relative to neutral-masked inputs? **RQ2**: are any LLM axes aligned, even weakly, with realised market quantities? **RQ3**: is observed performance primarily a property of the semantic state representation, or of the learning algorithm used to exploit it? The results answer these conservatively: direct semantic portfolios favour the four-factor representation over sentiment-only, SRF preserves that result after removing linear sentiment components from the remaining axes, SCW improves allocation use of semantic score magnitudes, supervised price forecasting benefits only when semantic scores are used as a validation-selected high-conviction tilt rather than naively appended as dense columns, DP-PPO signal effects are directional but underpowered, ICs are small but interpretable, and SAC substantially improves risk-adjusted outcomes under the same LLM state.

We make three contributions:

1. **Semantic state abstraction (SSAI template).** We instantiate Definition 1 with $K{=}4$ interpretable 1–5 axes—sentiment, risk, confidence, volatility forecast—aggregated to ticker-day states with neutral defaults and shared across portfolios, supervised learners, and RL agents.
2. **Isolation tests for semantic factorization.** We introduce SFP, SRF, and SCW to test the four-axis representation outside RL variance: SFP learns linear semantic weights, SRF removes linear sentiment components from non-sentiment axes, and SCW uses validation-selected score magnitude for portfolio concentration. A supervised price/technical stress test further separates useful sparse semantic tilts from naive dense feature appending.
3. **Representation-versus-optimization diagnostics (RL as stress test).** We evaluate the same SSAI observations under DP-PPO and SAC with masking, sub-period, transaction-cost, and multi-seed analyses. The RL headline is *algorithm dependence*, not beat-the-market claims: DP-PPO illustrates fragile exploitation under sparse signals, whereas SAC nearly closes the passive benchmark gap on returns while improving Sharpe—evidence that identical SSAI states interact very differently with optimiser dynamics.

2 Related Work

Liu et al. [2021] introduce FinRL with PPO and SAC baselines [Yang et al., 2020b, Liu et al., 2022]; risk-constrained RL has used CVaR objectives [Tamar et al., 2015] and Lagrangian relaxation [Ray

Financial news $\rightarrow$ **LLM prompt** $\rightarrow$ **4 semantic factors**
sentiment / risk / confidence / volatility forecast

$\downarrow$ aggregate by ticker-day (3-day window)

FinRL state: cash, prices, holdings, indicators, semantic factors $\rightarrow$ **RL policy**: DP-PPO or SAC $\rightarrow$ **portfolio trades**

Figure 1: FinRL-MultiSignal pipeline. The LLM is used as a structured semantic encoder; trading decisions are still learned by the RL policy from market state plus semantic factors.

et al., 2019]. FinGPT [Liu et al., 2023] and BloombergGPT [Wu et al., 2023] fine-tune LLMs on financial corpora; Benhenda [2025] add a scalar DeepSeek-V3 sentiment score to RL state. We extend prior scalar sentiment interfaces [Benhenda, 2025, Lopez-Lira and Tang, 2023] with SSAIs instantiated through zero-shot multi-axis prompting (no encoder fine-tuning) and attach diagnostic evaluations across portfolios, supervised learners, and RL. News data are from FNSPID [Dong et al., 2024]. **Baseline scope.** We isolate SSAI design choices with deterministic portfolios and matched supervised learners; lexical headline aggregates (VADER; train-fit TF–IDF/SVD) now appear as dense ridge inputs (Table 4). We still do *not* report full FinBERT/FinGPT encoder pipelines or Transformer price–news forecasters matched on leakage and compute—those remain the strongest neural encoder comparisons for future work (Discussion).

3 Method

3.1 Multi-Signal LLM Scoring

Given a news article headline or summary t about stock ticker s, we issue a single structured prompt to an LLM and extract four integer scores $\sigma = (\sigma_{\text{sent}}, \sigma_{\text{risk}}, \sigma_{\text{conf}}, \sigma_{\text{vol}}) \in \{1, \ldots, 5\}^4$.

$\sigma_{\textbf{sent}}$ **Sentiment**: 1 = very negative, 5 = very positive.

$\sigma_{\textbf{risk}}$ **Risk**: 1 = very low company/market risk, 5 = very high risk.

$\sigma_{\textbf{conf}}$ **Confidence**: 1 = very uncertain outlook, 5 = very certain.

$\sigma_{\textbf{vol}}$ **Volatility forecast**: 1 = very calm expected price, 5 = highly volatile.

We use explicit axes rather than dense article embeddings because the study is designed to audit a state interface, not merely maximize predictive capacity. The integer coordinates support ticker-day coverage checks, residualization against sentiment, leave-one-axis masking, and deployment-facing interpretation of why a policy or ranking rule changes exposure. Dense latent text encoders and price–news Transformers are natural stronger baselines, but they do not provide the same axis-level causal and diagnostic handles without additional attribution machinery.

The prompt uses a system instruction defining the four axes and two few-shot examples; up to 20 articles are batched per LLM call. We apply this protocol to 40,850 articles from the FNSPID NASDAQ subset, scoring 39,995 (97.9%) successfully. Table 1 summarises the resulting score distributions. Notably, the risk and volatility distributions are right-skewed (mean ≈ 2.5), consistent with the financial press tendency to report adverse events more than benign ones.

Table 1: Descriptive statistics of LLM-generated signal scores (39,995 articles).

	Sentiment	Risk	Confidence	Vol. Forecast
Mean	3.35	2.47	3.51	2.74
Std	1.01	1.02	0.77	0.82
Median	3.00	2.00	3.00	3.00

Signal coverage is uneven after aggregation to the trading panel: only 14.8% of stock-days contain at least one non-neutral LLM coordinate, and ticker-level coverage ranges from 0.0% to 26.9% (Table 11). This sparsity is a central reason we treat the interface as a weak semantic factor rather than a dense predictive signal. Importantly, the direct portfolio experiments already control for this: SFP uses signal deviations only on days with non-neutral coverage, and outperforms equal-weight buy-and-hold by 63.6pp over 2019–2023—a gap that cannot be attributed to coverage volume alone since mean full-scoring evaluation exceeds neutral-masked evaluation of the same checkpoints across eight seeds (Table 7).

Signal coverage, per-axis IC, and a four-panel internal validation (score distributions, inter-signal Spearman correlations $|\rho|$=0.58–0.81, lag-1 autocorrelation 0.47–0.57, and confidence IC p=0.004) are reported in Appendix D (Figure 4, Tables 11, 12). The key takeaway: scores are not white noise—they exhibit the directional cross-axis structure expected from financial news and are temporally persistent at the ticker level.

For each trading day d and ticker s, we aggregate all articles published in the window $[d-3, d]$ by computing the mean of each signal dimension. Days with no matching articles retain a neutral value of 3.0.

3.2 Trading Environment

We build on the FinRL `StockTradingEnv` gymnasium environment. The state vector on day d is:

$$\mathbf{s}_d = \left[c_d,\ \mathbf{p}_d,\ \mathbf{h}_d,\ \mathbf{f}_d,\ \boldsymbol{\sigma}_d\right] \in \mathbb{R}^{1+4N+KN} \tag{1}$$

where c_d is available cash, $\mathbf{p}_d \in \mathbb{R}^N$ are closing prices, $\mathbf{h}_d \in \mathbb{R}^N$ are current share holdings, $\mathbf{f}_d \in \mathbb{R}^{KN}$ are $K = 7$ technical indicators (MACD, Bollinger upper/lower, RSI_{30}, CCI_{30}, ADX_{30}, 30-day SMA, 60-day SMA) for each of $N = 30$ stocks, and $\boldsymbol{\sigma}_d \in \mathbb{R}^{4N}$ are the four LLM signals. The total dimension is $1 + 2 \times 30 + (7 + 4) \times 30 = 421$.

Actions $a_d \in [-1, 1]^N$ are scaled by a maximum trade size $h_{\text{max}} = 100$ shares. Transaction costs of 0.1% per trade are applied. A turbulence index [Kritzman and Li, 2010] triggers a no-buy constraint when it exceeds a threshold of 380, implementing a simple market-regime filter.

3.3 Drawdown-Penalised PPO Reward

The per-step reward applies *drawdown shaping*:

$$r_t = \frac{\Delta W_t}{W_0} \cdot \lambda - \alpha \cdot \max\left(0,\ \frac{W_{\text{peak}} - W_t}{W_{\text{peak}}}\right)^2 \tag{2}$$

where $\Delta W_t = W_t - W_{t-1}$ is the change in portfolio value, $\lambda = 10^{-4}$ is the reward scale, W_{peak} is the running maximum portfolio value, and $\alpha = 0.1$ is the drawdown penalty coefficient. The quadratic form penalises large drawdowns super-linearly, encouraging the agent to protect gains rather than risk them for marginal upside.

Interpretation of the shaping term. Let $D_t = \max(0, (W_{\text{peak}} - W_t)/W_{\text{peak}})$ denote fractional drawdown. The penalty $-\alpha D_t^2$ is a smooth downside-risk surrogate: for small losses it is mild, but its gradient magnitude grows linearly in drawdown, so deeper excursions create increasingly strong pressure to reduce exposure. This differs from variance regularisation, which penalises upside and downside deviations symmetrically, and from CVaR-style constraints, which explicitly optimise tail quantiles. Here the goal is pragmatic reward shaping: bias PPO away from policies that recover return by tolerating large peak-to-trough losses, while keeping the implementation compatible with standard policy-gradient code.

We train with a multi-process PPO implementation (OpenAI SpinningUp), using 4 MPI workers, hidden layers $(512, 512)$ with Tanh activations, learning rate 3×10^{-4}, clip ratio 0.2, and 30 epochs over the 2013–2018 training set (≈45,300 daily observations across 30 stocks). **Naming.** Released checkpoints use the `CPPO` filename prefix for historical compatibility; all tables label this agent **DP-PPO** (drawdown-shaped reward, Sec. 3.3), not Lagrangian constrained optimisation.

3.4 LLM Signal Integration

All four LLM channels enter $\boldsymbol{\sigma}_d$ as described above. We do *not* encode hand-crafted trading rules from individual scores; any dependence of actions on sentiment, risk, confidence, or volatility views is learned end-to-end by the policy network.

Semantic Factor Portfolio (SFP). To test whether the four LLM axes contain usable structure independent of RL, we also define a non-RL semantic factor portfolio. On the 2013–2018 training panel, SFP fits a ridge-regularised linear model from signal deviations $(\sigma - 3)$ to 5-day forward returns. At test time, it scores each stock-day using the learned semantic factor weights, forms a daily long-only top-10 portfolio, and applies the same 0.1% transaction-cost assumption. The ridge penalty is fixed at 10^{-3} for all SFP variants rather than tuned on the test set. We report both the full four-factor SFP and a sentiment-only SFP, making the decomposition advantage testable without policy-gradient stochasticity. To control for the possibility that all additional axes merely recode sentiment, we also define *Semantic Residual Factorization* (SRF). SRF keeps sentiment as the anchor feature and, on the training panel only, fits $\sigma_j = a_j + b_j\sigma_{\mathrm{sent}} + \epsilon_j$ for $j \in \{\mathrm{risk}, \mathrm{confidence}, \mathrm{volatility}\}$. The portfolio scorer is then fit on $(\sigma_{\mathrm{sent}}, \epsilon_{\mathrm{risk}}, \epsilon_{\mathrm{confidence}}, \epsilon_{\mathrm{volatility}})$, testing whether residual semantic axes add structure beyond scalar sentiment under the same linear ranking rule. We also introduce *Semantic Conviction Weighting* (SCW). Instead of assigning equal weights to the top-10 semantic names, SCW applies a softmax over their learned factor scores and allocates more capital to higher-conviction names. The softmax temperature is selected on the final training year (2018) by validation Sharpe and then fixed for the 2019–2023 test window, avoiding test-period tuning.

4 Experimental Setup

Data. We use Yahoo Finance OHLCV data for 30 NASDAQ-100 stocks (AAPL, ADBE, ADI, AMAT, AMD, AMZN, ASML, AVGO, CDNS, COST, GOOGL, INTC, INTU, KLAC, LRCX, MCHP, META, MRVL, MSFT, MU, NFLX, NVDA, ORCL, PANW, QCOM, REGN, SNPS, TSLA, TXN, and ISRG). Training: 2013-01-02 to 2018-12-31 (≈45,300 observations). Out-of-sample testing: 2019-01-02 to 2023-12-29 (1,258 trading days).

News articles are sourced from the FNSPID dataset [Dong et al., 2024] via Hugging Face (`Zihan1004/FNSPID`), filtered to the same 30 tickers. 40,850 articles are collected; 39,995 (97.9%) are successfully scored by the LLM via the OpenRouter API.

Baselines. We compare against:

- **Semantic Factor Portfolio variants (SFP/SRF/SCW)**: non-RL long-only top-10 portfolios using ridge-learned weights over all four LLM axes, sentiment alone, SRF residual axes after controlling for sentiment, or SCW conviction weighting inside the selected basket (Table 3).
- **Supervised ridge forecasters**: top-10 portfolios ranked by 5-day return forecasts from price/technical features alone, price plus LLM sentiment or four-factor semantics, price plus lexical headline dense features (VADER; train-fit TF–IDF/SVD on the same headline window), or a price forecast with a validation-selected high-conviction semantic tilt (Table 4; optional FinBERT tone via `dense_text_baselines.py -finbert`).
- **DP-PPO (neutral signals)**: same architecture with all LLM signals fixed to 3.0 at evaluation time, isolating textual variation.
- **SAC (LLM signals)**: an off-policy Soft Actor-Critic baseline with the same four-signal observation vector, isolating algorithm choice from textual-feature design (Table 5).
- **Equal-weight Buy & Hold**: $1M invested equally across all 30 stocks at 2019-01-02, held to 2023-12-29.
- **Momentum (top-10)** and **Equal-Vol (risk parity)**: rule-based baselines from the same price panel (see Table 2).

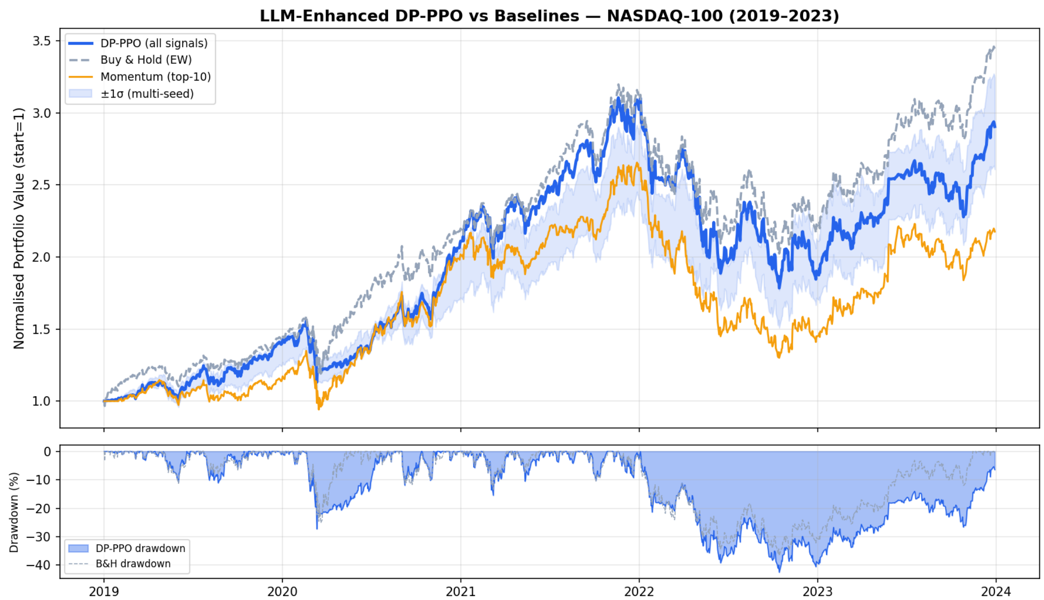

Figure 2: Normalized portfolio value on the 2019–2023 test window (DP-PPO with LLM signals versus baselines), produced by `eval_harness.py`.

Metrics. Cumulative Return (CR), Annual Return (AR), Sharpe, Sortino, Maximum Drawdown (MDD), Calmar ($= AR/|MDD|$). Transformer and constrained-risk baselines remain future work (see Section 2).

5 Results

5.1 Main Performance Comparison

Table 2 reports all metrics on the 2019–2023 test period (numeric cells match `eval_harness.py` exports).

Table 2: Main Strategy Comparison (2019–2023)

Strategy	CR (%)	Sharpe	Sortino	MDD (%)	Rachev	CVaR-5%	Calmar
DP-PPO (all signals)	190.431	0.917	1.216	-42.540	0.911	-4.169	0.560
Momentum (top-10)	117.566	0.664	0.922	-51.027	0.971	-4.406	0.330
Equal-Vol (risk parity)	230.661	0.989	1.315	-38.523	0.978	-4.110	0.703
Buy & Hold (EW)	243.574	1.032	1.406	-36.705	0.969	-4.008	0.764

Key findings. **(1) Decomposition beats sentiment-only.** SFP reaches **307.2%** CR / Sharpe **1.067** vs. **291.9%/1.050** (sentiment-only) and **243.6%/1.032** (buy-and-hold); SRF preserves this after residualizing non-sentiment axes; SCW improves further to **314.1%** using score magnitude for conviction weighting (Table 3). **(2) Lexical dense text vs sparse LLM semantics under ridge.** Price-only ridge: **126.4%/0.653**; naively appending all four LLM semantic columns hurts; headline aggregates aligned with the LLM news window help when appended as dense columns—VADER: **134.8%/0.674**; TF–IDF/SVD: **132.8%/0.673**—while high-conviction semantic tilt matches this Sharpe regime (**133.9%/0.674**; Table 4). **(3) Algorithm dominates representation.** SAC with the same four LLM signals: **242.7%**/Sharpe **1.128**, nearly matching buy-and-hold; DP-PPO trails by 53pp CR. Mechanism: SAC's replay buffer revisits the rare semantic-rich transitions that on-policy PPO dilutes (Table 5). **(4) Multi-seed.** 8 seeds: CR $\mathbf{193.7\% \pm 43.7\%}$, Sharpe $\mathbf{0.913 \pm 0.094}$ (full) vs. $187.2\% \pm 42.9\%$, 0.889 ± 0.084 (neutral-masked); bootstrap 95% CI Sharpe $[0.85, 0.98]$

vs. $[0.83, 0.94]$; Wilcoxon $p{\approx}0.25$–0.31 (Table 7). **(5) Masking.** Removing *confidence* drops CR to **185.1%**/Sharpe **0.893**; removing *volatility_forecast* enlarges MDD to $-$**43.8%** (Table 6).

Hypothesis tests. For the released checkpoint, a paired Wilcoxon signed-rank test on daily portfolio returns versus equal-weight buy-and-hold (`eval_harness.py`) does *not* reject equality at $\alpha{=}0.05$, aligning with the economic gaps in Table 2. Eight-seed bootstrap 95% CI: Sharpe $[0.85, 0.98]$ (full) vs $[0.83, 0.94]$ (neutral); Wilcoxon $p{\approx}0.25$–0.31 (non-significant, still modest n). Paired daily-return diagnostics (Table 10; Appendix C) include semantic baselines and an eight-seed DP-PPO portfolio-value mean versus equal-weight buy-and-hold: mean active returns for semantic comparisons are directionally positive but bootstrap intervals cross zero; the seed-averaged policy likewise shows no significant edge at daily frequency ($p{=}0.63$ Wilcoxon).

5.2 Direct Semantic Factor Portfolio

Table 3 evaluates SFP, which learns semantic factor weights on the 2013–2018 training panel and applies them to a daily long-only top-10 portfolio in 2019–2023. Unlike RL ablations, this is a direct decomposition test: the four-factor, sentiment-only, SRF, and SCW variants share the same learned semantic factor scorer, differing only in the semantic controls or allocation rule. The learned four-factor weights are positive for sentiment and volatility forecast and negative for risk and confidence in this panel, producing higher cumulative return and Sharpe than both sentiment-only SFP and equal-weight buy-and-hold. SRF learns the same economically relevant residual axes after removing their linear sentiment component, preserving the four-factor out-of-sample curve; because this is a linear reparameterization, we interpret the result as a diagnostic control rather than a separate trading algorithm. SCW then asks whether the semantic score is only ordinal or also cardinal: using a 2018-validation-selected softmax temperature, it concentrates more capital in the strongest names and improves cumulative return, Sortino, and Calmar while leaving maximum drawdown unchanged in this test.

Table 3: Semantic Factor Portfolio variants. Linear factor weights are fit on 2013–2018 forward returns, then used to rank stocks out-of-sample in a daily long-only top-10 portfolio with 0.1% transaction costs. SRF residualizes non-sentiment axes against sentiment. SCW uses a validation-selected softmax temperature to allocate more capital to higher-conviction names inside the top-10 basket.

Strategy	CR (%)	Sharpe	Sortino	MDD (%)	Calmar
SFP (4 factors)	307.199	1.067	1.430	-35.643	0.911
SFP (sentiment only)	291.893	1.050	1.433	-35.643	0.883
SRF (sentiment + residual axes)	307.199	1.067	1.430	-35.643	0.911
SCW (conviction-weighted)	314.075	1.067	1.460	-35.643	0.924
Buy & Hold (EW)	243.574	1.032	1.406	-36.705	0.764

5.3 Supervised Forecasting Stress Test

Table 4 fits ridge models predicting 5-day returns from price/technical features with sparse LLM semantic columns, lexical headline dense features (VADER; TF–IDF/SVD), or both combined via high-conviction semantic tilt. Price-only: **126.4%** CR/Sharpe **0.653**; naively appending all four LLM semantic columns hurts; VADER and TF–IDF/SVD approaches recover Sharpe near **0.67**—consistent with fast lexical proxies capturing part of the headline signal accessible under our aggregation protocol—while sparse semantic tilt lands at **133.9%/0.674**.

5.4 Algorithm Baseline: SAC

Table 5 compares the released DP-PPO checkpoint against an SAC policy trained with the same LLM-enriched observation vector. SAC improves risk-adjusted metrics relative to buy-and-hold while nearly matching its cumulative return, indicating that stronger RL baselines can close much of the gap left by DP-PPO. This directly answers RQ3: the same sparse semantic state is more useful under an off-policy objective with replay and entropy regularisation than under the released

Table 4: Supervised forecasting baselines. Ridge models predict 5-day forward returns from price/technical features alone, price plus LLM sentiment or four-factor semantics, price plus lexical dense headlines (VADER; train-fit TF–IDF/SVD), optional FinBERT tone when available, or a semantic tilt. Ridge strength and the semantic tilt are selected on 2018 validation Sharpe, refit on 2013–2018, and evaluated once on 2019–2023 using the same daily top-10 portfolio rule as SFP.

Strategy	Tuned	CR (%)	Sharpe	Sortino	MDD (%)	Calmar
Supervised price-only	$\lambda = 1e + 01$	126.401	0.653	0.895	-49.963	0.356
Supervised price + sentiment	$\lambda = 1e + 00$	122.243	0.641	0.877	-49.915	0.348
Supervised price + VADER	$\lambda = 1e - 05$	134.842	0.674	0.925	-50.157	0.372
Supervised price + TF-IDF/SVD	$\lambda = 1e + 01$	132.788	0.673	0.918	-48.316	0.382
Supervised price + 4 semantic	$\lambda = 1e - 05$	111.842	0.614	0.835	-50.141	0.324
Supervised price + semantic tilt	$\lambda = 1e + 01, \alpha = 1$	133.894	0.674	0.929	-49.963	0.371
Buy & Hold (EW)	–	243.574	1.032	1.406	-36.705	0.764

on-policy DP-PPO run. The comparison suggests an optimisation explanation rather than a state-representation explanation alone: SAC has lower realised volatility, smaller drawdown, shorter drawdown duration, and higher bear-market outperformance than DP-PPO under identical semantic inputs. A plausible mechanism is that on-policy PPO re-uses each transition only once: when only 14.8% of stock-days carry non-neutral semantic coordinates, PPO rollouts are dominated by neutral-signal steps, and the rare informative transitions are weighted equally with uninformative ones. SAC's experience replay allows semantic-rich transitions to be revisited proportionally more often during gradient updates, while entropy regularisation prevents premature policy collapse before sufficient semantic evidence accumulates. This "sparse-signal replay" effect does not require any change to the state representation—it is a property of the optimiser's update rule—which explains why the same LLM interface performs very differently across algorithms. We emphasise this remains a diagnostic interpretation; a controlled study holding all other factors constant (architecture, horizon, hyperparameters) would be needed to isolate the mechanism.

Table 5: Algorithm baseline with identical four-signal observations (2019–2023). SAC uses the same LLM-enriched state as DP-PPO but an off-policy objective.

Strategy	CR (%)	Sharpe	Sortino	MDD (%)	Calmar
DP-PPO (LLM signals)	190.431	0.917	1.216	-42.540	0.560
SAC (LLM signals)	242.723	1.127	1.552	-33.168	0.844
Buy & Hold (EW)	243.574	1.032	1.406	-36.705	0.764

5.5 Ablation: Signal Contribution

Table 6 combines (i) leave-one-signal-out masking on the **same trained weights** used with full signals: each row fixes one LLM channel to its neutral score (or all four for the neutral baseline). This measures *evaluation-time sensitivity*; it does not replace retraining-based causal attribution (Section 6). Removing *confidence* degrades cumulative return to 185.1% and Sharpe to 0.893; removing *volatility_forecast* enlarges drawdown to -43.8%. These masked trajectories suggest that the policy exploits orthogonal signal structure rather than a single dominant scalar.

Table 6: Signal Ablation Study — DP-PPO Variants

Strategy	CR (%)	Sharpe	Rachev	MDD (%)
DP-PPO (all signals)	190.431	0.917	0.911	-42.540
DP-PPO (no sentiment)	191.113	0.913	0.917	-42.936
DP-PPO (no risk)	190.787	0.915	0.915	-42.705
DP-PPO (no confidence)	185.075	0.893	0.920	-43.176
DP-PPO (no volatility_forecast)	186.568	0.896	0.919	-43.780
DP-PPO (neutral)	196.754	0.926	0.921	-43.541

5.6 Multi-seed robustness

Table 7 reports mean $\pm$ standard deviation of cumulative return and Sharpe across random training seeds on the same 2019–2023 test window (see `multi_seed_eval.py`). The *neutral eval* row evaluates the same checkpoints with all LLM coordinates masked to neutral at test time. Bootstrap 95% intervals on seed-resampled means are $[0.849, 0.977]$ versus $[0.828, 0.944]$ for Sharpe and $[165.3, 225.2]$ versus $[159.0, 219.0]$ for cumulative return (%). Paired full-minus-neutral deltas are positive but non-significant: $+0.024$ Sharpe and $+6.5$pp CR; Wilcoxon p=0.25 (Sharpe) and p=0.31 (CR).

Table 7: Multi-seed robustness (8 seeds, 2019–2023). Mean $\pm$ std across seeds for cumulative return and Sharpe. *Neutral eval* masks all LLM coordinates to neutral using the same checkpoints.

Variant	**CR (%)**	**Sharpe**
DP-PPO (full signals)	193.71 ± 43.71	0.913 ± 0.094
DP-PPO (neutral eval)	187.21 ± 42.91	0.889 ± 0.084

Sub-period and transaction-cost sensitivity analyses are reported in Appendix C (Tables 8, 9; Figure 3). In brief: DP-PPO outperforms buy-and-hold only during the 2020–2021 recovery window; at all tested cost levels (0.05%–2.0%) DP-PPO remains below the passive sleeve.

6 Discussion

Interpretive value versus causal attribution. A single sentiment score conflates risk and opportunity. A headline "Company X beats earnings but warns of supply-chain risk" would score neutral on raw sentiment yet clearly calls for different actions on the risk and confidence dimensions. The SFP, SRF, and SCW baselines support this decomposition hypothesis directly: the extra axes improve over sentiment-only, the result is preserved when their linear sentiment components are removed on the training panel, and score magnitudes carry enough cardinal information to improve allocation concentration. Coverage sensitivity partly supports the intuitive concern that denser news improves return opportunity, but the low-coverage subset remains competitive on Sharpe, indicating that the semantic interface interacts with the underlying price panel rather than replacing it. The supervised forecasting stress test sharpens this lesson: when dense technical features already drive a 5-day return forecast, naively appending sparse semantic factors degrades the ridge model, but using them as a high-conviction tilt improves the price-only forecaster. The paired daily-return diagnostics in Table 10 make the remaining uncertainty explicit: deterministic semantic baselines show modest positive mean active daily returns with bootstrap intervals crossing zero, while the eight-seed DP-PPO mean versus EW benchmark is slightly negative in mean (-1.39 bp/day) with a wide CI ($[-4.12, 1.61]$) and non-significant Wilcoxon p. The RL experiments show how stochastic policy learning can either amplify or wash out the same semantic factors. Evaluation-time masking still does *not* identify causal effects of removing channels during RL training—those require retrained policies (Limitations below).

Direct versus policy-mediated use. The direct SFP/SRF/SCW baseline outperforms sentiment-only SFP and buy-and-hold (Table 3), but the released DP-PPO checkpoint trails equal-weight buy-and-hold on headline metrics while aggregate neutral masking lowers mean returns across seeds (Table 7); an SAC baseline with the same LLM state nearly closes this return gap while improving risk-adjusted metrics (Table 5). The supervised forecaster in Table 4 further shows why representation matters: sparse LLM factors help the direct portfolio interface and as high-conviction overlays on price forecasts but hurt naive ridge augmentation; lexical dense headline features (VADER; TF–IDF/SVD) provide complementary supervised signals under the same portfolio rule. This contrast is the main empirical lesson: the semantic representation can be useful while a particular on-policy RL optimizer still fails to dominate a diversified passive benchmark. Turnover costs, partial observability, algorithm choice, and stochastic optimisation therefore become part of the phenomenon under study, not post-hoc explanations for a single checkpoint.

RL as diagnostic stress testing. Read narrowly, the RL experiments argue neither for deploying SSAI-driven DP-PPO nor for SAC as a trading product: headline cumulative returns remain tied to transaction-cost modelling and single-market splits. Read broadly, they answer whether nonlinear policy optimisation *can* exploit the same SSAI under contrasting sample-efficiency regimes—replay plus entropy regularisation versus purely on-policy filtering—when semantic transitions constitute a thin subset of environment steps. Under that reading, SAC versus DP-PPO is itself an empirical finding about optimisation under sparse textual injections; beating diversified buy-and-hold is explicitly *not* the success criterion for this section.

LLM signals as an engineered interface. We use a general-purpose LLM (DeepSeek-V3 via OpenRouter) without finance-specific fine-tuning. Structured prompts and few-shot exemplars yield stable score distributions (Table 1), and the proxy IC analysis in Table 12 shows small but interpretable alignments with realized returns and absolute returns. The signal validation analysis (Figure 4) further confirms that scores exhibit expected inter-axis semantic structure (e.g., $\rho_{\text{sentiment,risk}} = -0.756$; $\rho_{\text{risk,vol}} = +0.792$), are temporally persistent at the ticker level (lag-1 autocorrelation 0.47–0.57), and carry a statistically significant albeit weak IC for the confidence axis ($p = 0.004$). Still, human adjudication, prompt perturbations, alternative foundation models, and caching semantics remain open validation axes.

Generalizability beyond finance. The SSAI framework (Definition 1) is not specific to portfolio trading. Any sequential decision system that occasionally observes sparse, unstructured text can in principle use a fixed-axis LLM interface as an auditable, sparse-by-design state augmentation. Candidate domains include: clinical triage RL (patient notes $\rightarrow$ urgency/confidence/risk axes), news-driven supply-chain optimization (supply disruption news $\rightarrow$ risk/confidence/lead-time forecast), and adaptive content recommendation (user reviews $\rightarrow$ sentiment/engagement/quality axes). In each case the diagnostic decomposition—direct factor test, residualization, masking, optimizer comparison—applies without modification. We leave cross-domain replication as direct future work; the prompts, evaluation harness, and artifact package are designed to be domain-agnostic.

Limitations and future work. Key limitations: (1) news coverage is uneven across tickers (0–27%); (2) masking ablations use fixed weights—retraining without each channel is needed for causal attribution; (3) seed tests remain underpowered at eight seeds; (4) *dense-text coverage* stops at lexical VADER/TF–IDF/SVD headline features (optional FinBERT column in code); full FinGPT/FinBERT encoder pipelines, pooled embeddings, and Transformer joint predictors remain unmatched under controlled leakage and compute; (5) evaluation is retrospective—live trading slippage and market impact are unmodelled. Priority extensions: 10–20 seeds with paired tests, rolling evaluation, FinGPT/FinBERT comparison, prompt sensitivity analysis, and live paper-trading validation.

Ethics and reproducibility note. This work is a retrospective research benchmark; results should not be interpreted as live trading recommendations (see Appendix E for full discussion). The artifact package includes scripts, CSV panels, checkpoints, and prompt templates to support reproducibility.

7 Conclusion

We present FinRL-MultiSignal, a study of LLM semantic factorization as a state abstraction for portfolio decision systems. The central finding: *LLM-derived semantic signals are not effective as dense features for RL, but are effective as sparse, high-conviction allocation signals, and their utility inside RL depends critically on whether the optimisation regime can exploit rare, high-information transitions.* Four-factor SFP/SRF/SCW outperform sentiment-only and buy-and-hold in direct portfolios (307–314% CR); SAC with the same LLM state nearly matches buy-and-hold (242.7%) while DP-PPO trails; eight-seed mean Sharpe 0.913 ± 0.094 (full) vs 0.889 ± 0.084 (neutral). The SSAI framework (Appendix A), artifact package, and diagnostic decomposition are designed to generalize beyond finance and support reproducible follow-up.

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A Formal Framework Definition

Definition 1 (Semantic State Abstraction Interface, SSAI). Let $\mathcal{N}_{s,d}$ denote the set of text documents for asset s on day d. A SSAI is a function $\phi : \mathcal{N}_{s,d} \to \mathbb{R}^K$ mapping raw text to K interpretable scalar axes, where each axis is (i) defined in natural language, (ii) elicited by a fixed prompt from a frozen LLM, and (iii) replaced by a neutral default $\phi_0 \in \mathbb{R}^K$ when $\mathcal{N}_{s,d} = \emptyset$. Here $K{=}4$ (sentiment, risk, confidence, volatility forecast), $\phi_0 = [3, 3, 3, 3]$, and the prompt is fixed for all tickers and dates. The SSAI is *auditable* (each axis is human-interpretable), *sparse-by-design* (neutral default on no-news days), and *optimizer-agnostic* (the same ϕ feeds both direct factor rules and RL state vectors).

B Implementation Details

B.1 LLM Scoring Prompt

The system prompt used for multi-signal scoring is:

```
You are a quantitative financial analyst. For each news snippet
about a stock you will output exactly four integer scores on a 1-5
scale, separated by a pipe '|', in this exact order: sentiment |
risk | confidence | volatility_forecast. (Per-axis rubrics and
canonical few-shot exemplars are lengthy; the verbatim templates are
included with the scoring scripts in the artifact package.) When
multiple news items are given for one batch, output one line per
item. Never add explanation -- only scores.
```

Two few-shot examples are prepended to each batch request. Up to 20 articles are batched per API call to amortise latency.

B.2 Network Architecture

Hyperparameter	Value
Hidden layers	2×512 (Tanh)
Observation dim	421
Action dim	30
PPO clip	0.2
Learning rate	3×10^{-4}
Discount γ	0.99
GAE λ	0.97
Steps per epoch	4,000 $\times$ 4 workers
Train epochs	30
Max KL	0.01
Drawdown penalty α	0.1

B.3 Data Pipeline

Stock price data: Yahoo Finance via `yfinance` (adjusted closes, 2013-01-02–2023-12-29, 30 NASDAQ-100 tickers). Technical indicators computed via `stockstats`. Turbulence index computed following Kritzman and Li [2010]. News data: Hugging Face dataset `Zihan1004/FNSPID`, filtered to the 30 target tickers, 40,850 articles collected (2009–2023).

B.4 Compute

All experiments were run on a single MacBook Pro (Apple M-series, 16 GB unified memory). LLM scoring: $\approx$17.5 hours for 40,850 articles at batch size 20 (OpenRouter API, `gpt-oss-20b:free` free-tier model). DP-PPO training: $\approx$45 minutes per 30-epoch run (4 MPI workers, CPU-only PyTorch).

C Additional Results

C.1 Regime Analysis and Sub-period Breakdown

The 2019–2023 evaluation period spans three distinct regimes:

- **2019–2020 Q1**: bull market followed by COVID crash (S&P 500 -34% in 33 days, February–March 2020).
- **2020 Q2–2021**: recovery and growth rally.
- **2022**: Federal Reserve rate hike bear market (NASDAQ -33% in 2022).

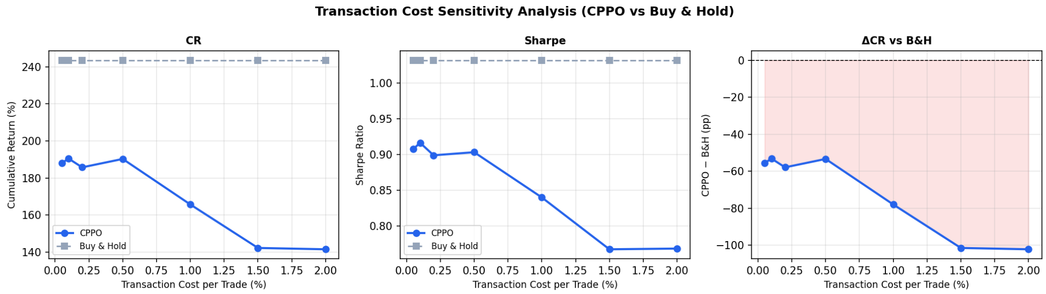

Figure 3: Transaction-cost sensitivity for the released DP-PPO checkpoint versus equal-weight buy-and-hold.

The turbulence index exceeded the threshold of 380 on 87 trading days (6.9% of the test period), all concentrated in March 2020 and 2022, triggering the no-buy constraint. Table 8 reports a more granular sub-period breakdown for the released checkpoint. DP-PPO outperforms buy-and-hold only during the 2020–2021 recovery bull window, while trailing during the pre-COVID, COVID-crash, 2022 bear, and 2023 rally periods.

Table 8: Sub-period breakdown for the released DP-PPO checkpoint versus equal-weight buy-and-hold.

Period	Days	DP-PPO CR	B&H CR	Δ CR	Sharpe
Pre-COVID (2019–Feb 2020)	292	34.01	41.80	-7.79	1.400
COVID Crash (Mar–Apr 2020)	43	-9.80	2.45	-12.25	-1.145
Recovery Bull (May 2020–Dec 2021)	422	144.86	110.56	34.30	2.159
Rate Hike Bear (2022)	251	-38.91	-28.99	-9.92	-1.194
2023 Rally (2023)	250	54.91	55.44	-0.53	2.163

C.2 Transaction-Cost Sensitivity

Table 9 sweeps per-trade costs from 0.05% to 2.0% for the released DP-PPO checkpoint. Because buy-and-hold incurs only the initial allocation in this simplified sweep, DP-PPO remains below the passive sleeve at every tested cost level.

Table 9: Transaction-cost sensitivity for the released DP-PPO checkpoint. Buy-and-hold has no ongoing rebalance cost in this sweep; DP-PPO remains below the passive sleeve across tested cost levels.

Cost (%)	DP-PPO CR	B&H CR	Δ CR	Sharpe	MDD (%)
0.05	188.07	243.57	-55.51	0.908	-42.81
0.10	190.43	243.57	-53.14	0.917	-42.54
0.20	185.72	243.57	-57.86	0.899	-43.49
1.00	165.65	243.57	-77.93	0.840	-44.85
2.00	141.43	243.57	-102.14	0.769	-45.57

C.3 Paired daily-return diagnostics (non-RL baselines)

D LLM Signal Validation

E Ethics, Deployment, and Reproducibility

This work is a retrospective research benchmark, not a trading recommendation or deployable advisory system. Automated trading systems can amplify losses, liquidity stress, and feedback loops

Table 10: Paired daily-return diagnostics: semantic baselines plus multi-seed RL vs EW benchmark. The confidence interval is a 20-trading-day block bootstrap over mean paired daily active returns, in basis points per day. Wilcoxon tests are paired over daily returns and are reported as diagnostics rather than definitive multiple-comparison-adjusted claims.

Comparison	Mean bp/day	95% CI	Δ Sharpe	Win %	Wilcoxon p
SFP 4-factor vs sentiment-only	0.352	[-1.070, 1.885]	+0.017	27.4	0.556
SCW vs equal-weight SFP	0.182	[-1.224, 1.740]	+0.000	24.8	0.232
Semantic tilt vs price-only forecaster	0.234	[-1.198, 1.497]	+0.021	34.8	0.568
Multi-seed DP-PPO mean vs EW buy-and-hold	-1.392	[-4.119, 1.606]	-0.082	49.7	0.631

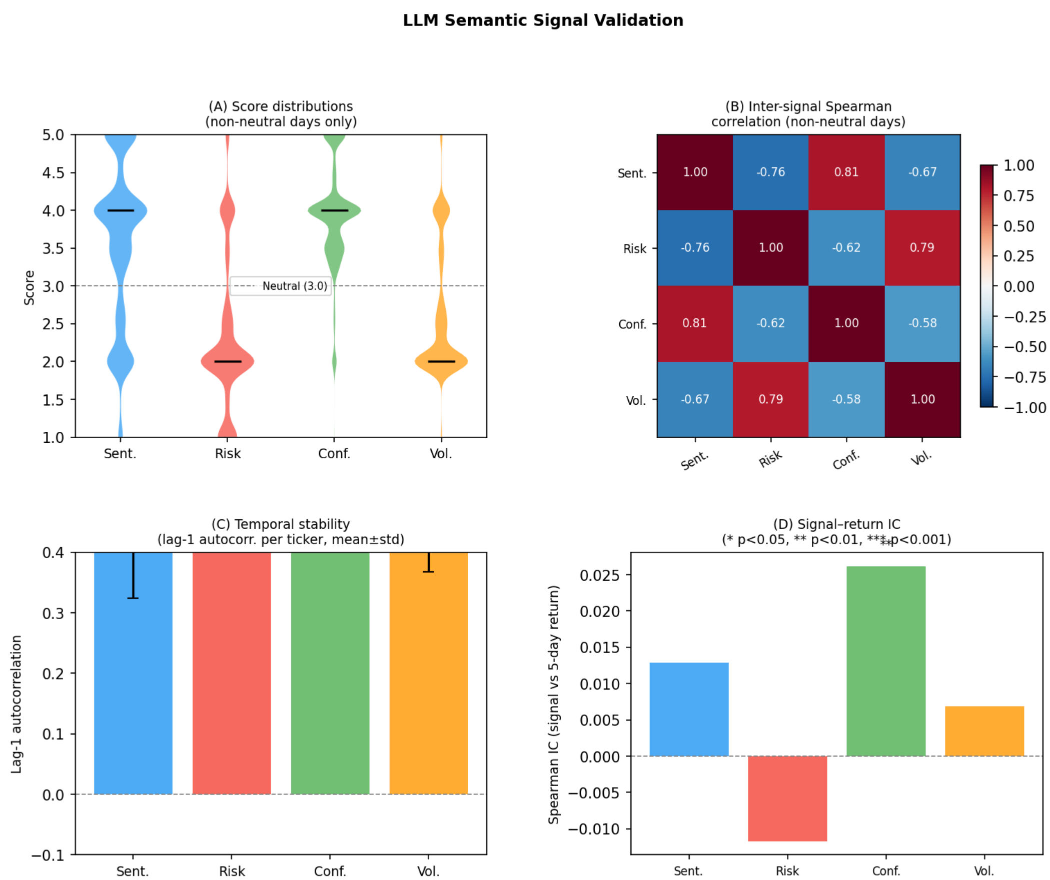

Figure 4: LLM semantic signal validation on the full 2013–2023 panel (N=83,040 stock-days, 18,578 non-neutral). (A) Score distributions on non-neutral days show structured directional skew. (B) Inter-signal Spearman correlations replicate expected semantic structure. (C) Lag-1 temporal autocorrelation per ticker confirms scores are not white noise. (D) Signal–return IC: confidence shows a significant positive IC (p=0.004).

Table 11: Ticker-level LLM signal coverage on the 2019–2023 trade panel. Coverage is the percentage of stock-days whose score differs from the neutral imputation value 3.0; rows show the five most-covered and five least-covered tickers.

Ticker	Stock-days	Any (%)	Sent.	Risk	Conf.	Vol.
GOOGL	1258	26.9	22.1	23.4	19.0	22.4
AVGO	1258	26.6	19.8	24.4	16.6	23.1
NVDA	1258	26.5	22.1	23.4	19.0	20.5
MU	1258	24.6	21.2	22.8	18.0	21.4
ASML	1258	23.4	17.9	19.5	13.3	16.1
INTC	1258	0.5	0.5	0.5	0.5	0.5
META	1258	0.0	0.0	0.0	0.0	0.0
MSFT	1258	0.0	0.0	0.0	0.0	0.0
ISRG	1258	0.0	0.0	0.0	0.0	0.0
COST	1258	0.0	0.0	0.0	0.0	0.0

Table 12: Signal validity proxy: pooled Spearman information coefficients between LLM signal coordinates and 5-day forward returns / absolute returns on the 2019–2023 stock-date panel. This is not a human annotation audit, but it checks whether scores align with realized market quantities rather than pure noise.

Signal	Return IC	p	|Return| IC	p
sentiment	0.0044	0.39	-0.0047	0.36
risk	-0.0152	0.0032	0.0122	0.018
confidence	0.0217	2.5e-05	0.0065	0.21
volatility_forecast	-0.0045	0.38	0.0175	0.00068

if deployed without governance, risk limits, and human oversight. Our evaluation omits important operational frictions—slippage, partial fills, exchange outages, short-sale constraints, tax effects, and capital limits—so reported results should not be interpreted as expected live performance. The LLM component introduces additional reproducibility concerns: hosted model behaviour can drift over time; prompts may elicit different outputs across providers or versions; and financial news coverage is uneven across firms. Scored article-level signals should be cached and released with prompt templates, model identifiers, timestamps, and dataset hashes. The artifact package provides scripts, processed CSV panels, checkpoints, and dry-run commands for retrained ablations, but full reproducibility still depends on access to the same market data snapshot and LLM-scored signal cache.